ISLAMIC UNIVERSITY OF TECHNOLOGY

Department of Computer Science and Engineering (CSE)
Course Outline and Course Plan

Name of the Teacher	**Njayou Youssouf**	**Position**	**Lecturer**
Department	**CSE**	**Program**	**BSc Engg. SWE**
Course Code	**CSE 4501**	**Course Title**	**Operating Systems**
Academic Year	**2025-2026**	**Semester**	**Winter**
Contact Hours	**3.0**	**Credit Hours**	**3.0**
Textbooks and Reference books	**1. Operating System Concepts 10th Edition** **2. Modern Operating Systems 5th Edition** **3. Operating Systems: Three Easy Pieces**	**Authors**	**1. Abraham Silberschatz, Greg Gagne, and Peter B. Galvin** **2. Andrew S. Tanenbaum and Herbert Bos** **3. Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau**
Prerequisites (If any)	**1. CSE 4303: Data Structures** **2. CSE 4305: Computer Organization and Architecture**		
Course Homepage	**https://classroom.google.com/u/1/c/ODkwNTg5MjA3NDA3**		
Teaching Methods/ Approaches	**☐ Lecture√**	**☐ Group discussion√**	**☐ Demonstration√**	**☐ Problem-solving√**
	☐ Project	**☐ Others: Programming assignments√**		
Teaching aids	**Multi-media√**	**OHP**	**Board and Marker√**	**Others**

Course Assessment Method						
Attendance (10%)	**Quizzes and Assignments: 15% of Total Marks**				**Mid Semester (25%)**	**Semester Final (50%)**
Class participation	**Assignment 1**	**Assignment 2**	**Assignment 3**	**Assignment 4**	**Week/Date**	**Week/Date**
	Unix shell (6%)	**CPU scheduler simulator (4%)**	**Banker's algorithm (2.5%)**	**Page replacement (2.5%)**		
	Set week 3, due week 7	**Set week 4, due week 6**	**Set week 6, due week 10**	**Set week 10, due week 12**	**As per the schedule of IUT**	**As per the schedule of IUT**

Grading Policy					
Marks out of 100	**Letter Grade**	**Grade Point**	**Marks out of 100**	**Letter Grade**	**Grade Point**
80 - 100	A+	4.00	55 - 59	B-	2.75
75 - 79	A	3.75	50 - 54	C+	2.50
70 - 74	A-	3.50	45 - 49	C	2.25
65 - 69	B+	3.25	40 - 44	D	2.00
60 - 64	B	3.00	00 - 39	F	0.00

Class Schedule	
Monday	9:15 AM - 10:30 AM
Thursday	11:45 AM - 1PM

Student's consulting hour	**Wednesday , 2:30 PM**

Course Contents
Types of operating systems: single user, real-time, batch, multiple access. Principles of operating systems; design objectives; sequential processes; concurrent processes, concurrency, functional mutual exclusion, processor co-operation and deadlocks, processor management. Control and scheduling of large information processing systems. Resource allocation, dispatching, processor access methods, job control languages. Memory management, memory addressing, paging and store multiplexing. Multiprocessing and time sharing, batch processing. Scheduling algorithms, file systems, protection and security; design and implementation methodology, performance evaluations and case studies.

Course Objectives
After completing the course, the student must be able to: Explain what operating systems do and how they are structured and designed. Explain how processes and threads are created, scheduled and synchronized, and how deadlocks are handled. Explain how operating systems manage main memory, virtual memory, file systems and I/O devices. Identify the protection and security mechanisms that operating systems provide. Describe multiprocessor, distributed and modern operating-system designs.

Mapping with CO, PO, and Bloom's Taxonomy			
CO No.	**Course Outcomes (CO) Statement**	**levels of Bloom's Taxonomy**	**Matching with Program Outcome (PO)**
CO1	Describe the evolution, types, structure, and functions of operating systems	C1	PO1
CO2	Analyze techniques involved in managing various operating system resources.	C2	PO1
CO3	Apply operating systems techniques to solve modern operating systems problems.	C3	PO2
CO4	Design solutions for effective resource management and system robustness.	C3	PO3

Weekly plan for course content and mapping with CO		
Weeks	**Topics**	**COs**

1	Introduction to Operating Systems (with class orientation)	CO1
2	OS Structure and Design Principles	CO1
3	Processes and Threads	CO1, CO4
4	CPU Scheduling	CO2
5	Process Synchronization	CO3
6	Deadlocks	CO2, CO3
7	Memory Management Basics	CO2, CO4
8 & 9	**MID-TERM EXAMINATIONS**	
10	Memory Management Advanced	CO2, CO4
11	File Systems	CO4
12	I/O Systems and Device Management	CO1
13	Protection and Security	CO4
14	Multiprocessing and Distributed Systems	CO1
15	Modern Operating Systems and Trends	CO1
16 - 19	**FINAL EXAMINATIONS**	

Mapping of Course Outcomes (COs), Program Outcomes (POs), and Evaluation Methods						
Assessment Method	**Marks**	Mark distributions (as %) on COs and POs				
		CO1	CO2	CO3	CO4	CO4
		PO1	PO1	PO2	PO3	PO3
Attendance (Class Participation)	10%	-	-	-	-	-
Assignment 1/Assignment 2/Assignment 3/Assignment 4	30%	-	4%	2.5%	6%	2.5%
Midterm Exam.	30%	10%	10%	5%	-	-
Final Exam.	30%	10%	15%	10%	-	15%
Total	100%	20%	29%	17.5%	6%	17.5%

Mapping of COs and POs [Correlation level 1 for low, 2 for moderate, and 3 for high]												
Course Outcomes	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3											
CO2	3											
CO3		3										
CO4			3									
CO5		3										

Program Outcomes	
PO 1	**Engineering Knowledge:** Apply knowledge of mathematics, natural science, engineering fundamentals and system fundamentals, software development, networking & communication, and information assurance & security to the solution of complex engineering problems in computer science and engineering.
PO 2	**Problem Analysis:** Ability to identify, formulate and analyze complex Computer Science and Engineering problems in the areas of hardware, software, theoretical Computer Science and applications to reach significant conclusions by applying Mathematics, Natural sciences, Computer Science and Engineering principles.
PO 3	**Design/ Development of Solutions:** Design solutions for complex computer science and engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
PO 4	**Investigation:** Ability to use research-based knowledge and research methods to perform literature survey, design experiments for complex problems in designing, developing and maintaining a computing system, collect data from the experimental outcome, analyze and interpret valid/interesting patterns and conclusions from the data points.
PO 5	**Modern Tool Usage:** Ability to create, select and apply state of the art tools and techniques in designing, developing and testing a computing system or its component.
PO 6	**The Engineer and Society:** Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice in system development and solutions to complex engineering problems related to system fundamentals, software development, networking & communication, and information assurance & security.
PO 7	**Environment and Sustainability:** Assess how computing solutions affect society and the environment, and apply the principles of sustainable development.
PO 8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of computer science and engineering practice.
PO 9	**Individual Work and Teamwork:** Ability to function as an individual and as a team player or leader in multidisciplinary teams and strive towards achieving a common goal.
PO 10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO 11	**Project Management and Finance:** Demonstrate knowledge and understanding of engineering management principles and economic decision making and apply these to one's own work, as a member and leader in a team to manage projects and in multidisciplinary environments.
PO 12	**Life-long learning:** Recognize the need for and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

Instructors contact details:

Njayou Youssouf
Lecturer
Department of Computer Science and Engineering
Islamic University of Technology
Room No: 502 2nd Academic Building.
Mobile: 01793570834
Email: njayou@iut-dhaka.edu